# Every 0/1 polytope has Chvatal–Gomory rank zero

Conjecture 00000003973

zbw123456 18 October 2026

Use the standard meaning of a 0/1 polytope: $P = \mathrm{conv}(V)$ for $V \subseteq \{0,1\}^n$. Its integer hull is $P_I = \mathrm{conv}(P \cap \mathbb{Z}^n)$. Since every vertex in $V$ is integral and $P$ is convex,

$$P = \mathrm{conv}(V) \subseteq \mathrm{conv}(P \cap \mathbb{Z}^n) \subseteq P.$$

Therefore $P_I = P$.

The CG closure is the intersection of $P$ with all cuts $a(x) \le \lfloor b \rfloor$, where $a$ is a real linear functional integer-valued on $\mathbb{Z}^n$ and $a(y) \le b$ for all $y \in P$. These are exactly integer-coefficient CG cuts. The CG rank is the least nonnegative $k$ for which $\mathrm{CG}^k(P) = P_I$. The zeroth iterate is $P$, so the equality already holds at $k = 0$. Hence every 0/1 polytope has rank zero.

In particular, no family of such polytopes, including partition polytopes in this class, can have ranks at least $cn^2$ for a fixed $c > 0$ and all sufficiently large $n$. Taking a positive $n$ above the threshold would give $0 \ge cn^2 > 0$. The upper-bound clause alone is trivial, but the quadratic-order attainment and $\Theta(n^2)$ supremum clauses are false. In every positive dimension the class is nonempty (a singleton integer vertex suffices), and its set of ranks is $\{0\}$.

**Scope.** A polytope with fractional vertices merely contained in $[0,1]^n$ is a different class. The proof uses the standard vertex-based term *0/1 polytope* present in both language versions and makes no claim about arbitrary relaxations inside the cube.

**Lean correspondence.** Lean defines the integer lattice, real convex hull, integer hull, CG closure, and rank as an `IsLeast` statement. `integerHull_eq` explicitly constructs integer coordinates for 0/1 vertices and proves both inclusions. `rank_zero` and `no_positive_rank` prove the exact rank. `no_quadratic_family` rules out every positive eventual quadratic lower bound. No rank or integer-hull assertion is assumed.

**Reproduction.** The standalone project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409` Run `sh verify.sh`; the Lean source prints the main results' axiom dependencies. No proof placeholders, native evaluator bypass, or custom axioms are used.

**Source.** The official bilingual statement is included as `conjecture.md`, checked at repository commit `cce2d90cb298de9595798eb674d2c80712100532`.